

Guangrui Fan

Date of Birth: July 1992 Location: Taiyuan, Shanxi, China Mobile: +86 13387574446

Email: gerryfan2022@gmail.com Homepage: gerryfan0706.github.io

ORCID: 0009-0001-8570-1636 GitHub: github.com/gerryfan0706

[Google Scholar Profile](#)

Research Interests

Human-AI Interaction, Social Computing, and Trustworthy Artificial Intelligence. My research examines the theories, methods, and technologies that enable intelligent systems to develop alongside individuals, groups, and social systems. My long-term goal is to integrate cognitive and behavioral modeling, machine learning, and interaction design to advance intelligent technologies and their trustworthy use in complex social environments.

- **Human-AI Interaction and Collaborative Intelligence:** Interaction mechanisms, collaboration models, and human capability augmentation, including human-AI collaboration in cognition, decision-making, and creativity.
- **Social Computing and Social Intelligence:** Individual and group behavior, social networks, information diffusion, and the evolution of mixed human-AI groups; modeling, analysis, and intelligent decision-making for complex social systems.
- **Trustworthy AI and Large Language Models:** Evaluation, alignment, reliability, and governance of LLMs and agents; intelligent systems that balance performance, fairness, privacy, and controllability.
- **Affective Computing and Social Cognition:** Computational modeling, expression, and interaction mechanisms for affect and social cognition; intelligent systems with contextual understanding and social adaptability.

Keywords: Human-AI Interaction; Collaborative Intelligence; Social Computing; Trustworthy AI; LLMs and Agents; Affective Computing.

Research Highlights

- **Research output:** This concise CV lists 23 published or accepted conference and journal papers as first or corresponding author, including 12 CCF-A papers (11 CCF-A conference papers and 1 CCF-A journal paper, *ACM TOIS*) and 5 CAS Zone 1 Top journal papers. One paper received an ACM CHI 2026 Honorable Mention Award; one paper was recognized as an ESI Highly Cited Paper.

Education

B.S. in Digital Media Technology Sept. 2010 – June 2014
School of Software, Huazhong University of Science and Technology, Wuhan, China [QS 2027: 307](#)

M.S. in Software Engineering Sept. 2014 – Feb. 2018
School of Software, Shanghai Jiao Tong University, Shanghai, China [QS 2027: 36](#)
Recommended admission without entrance examination

Ph.D. in Computer Science Sept. 2022 – June 2026
Universiti Malaya, Kuala Lumpur, Malaysia [QS 2027: 56](#)
Research areas: Human-AI Interaction, Social Computing, Affective Computing

Professional Experience

Lecturer June 2018 – Sept. 2022
School of Computer Science and Technology, Taiyuan University of Science and Technology, Taiyuan, China

Publications

Note: * corresponding author; # co-first author; my name is bolded. CCF ranks follow the [2026 directory](#); IJCAI 2025 retains its CCF-A rank at publication. CSCW and UbiComp are counted by conference rank. The A* note for Findings refers only to the EMNLP main conference's ICORE rank.

Conference Papers

CCFA CCF-A Conferences (11 papers)

[C1] **CCFA** Liu, D., & **Fan, G.*** (2027). When the Score Speaks First: Negotiating the Authority of Wearable Wellness Scores in Everyday Life. *ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC 2027)*. **Accepted; forthcoming.**

Category: CCF-A conference (UbiComp); Role: Corresponding author.

[C2] **CCFA** Liu, D., Sabri, A.Q.M., Rahman, S.S.A., Pan, L., & **Fan, G.*** (2027). When AI Breaks Down: Continuity Work, Dependency, and Repair in AI-Mediated Collaboration. *ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2027)*. **Accepted; forthcoming.**

Category: CCF-A conference (CSCW); Role: Corresponding author. Accepted 14 September 2026; forthcoming.

- [C3] **CCFA** Fan, G.* , Liu, D., Sabri, A.Q.M., Zhang, R., & Pan, L. (2026). Feeling Rules in Language Models: Mapping Norms of Emotional Appropriateness Across Roles, Institutions, and Intensity. *In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*. pp. 10175–10193. ACL. [ACL Anthology](#).
Category: CCF-A conference; top-tier venue in natural language processing.
- [C4] **CCFA** Fan, G.* , Liu, D., Sabri, A.Q.M., & Pan, L. (2026). Graph-Preference Learning: Debiasing Network-Sampled Human Feedback for Target Welfare Estimation. *In Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*. PMLR. [Conference listing](#).
Category: CCF-A conference; top-tier venue in machine learning.
- [C5] **CCFA** Fan, G.* , Liu, D., & Pan, L. (2026). Is It Still You? Attributing Authorship and Authenticity in AI-Assisted Romantic Communication. *In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI 2026)*, ACM. DOI: [10.1145/3772318.3790762](#).
Category: CCF-A conference; top-tier venue in HCI.
- [C6] **CCFA** Fan, G.* , Liu, D., Pan, L., & Zhang, R. (2026). When Help Hurts: Verification Load and Fatigue with AI Coding Assistants. *In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI 2026)*, ACM. DOI: [10.1145/3772318.3791176](#).
Category: CCF-A conference; top-tier venue in HCI. **Recognition**: ACM CHI 2026 Honorable Mention Award.
- [C7] **CCFA** Fan, G.* , Liu, D., & Pan, L. (2026). Co-Adaptive Eco-Nudging: A Privacy-Preserving Contextual Bandit with User-Taught Preferences in Everyday Browsing. *In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI 2026)*, ACM. DOI: [10.1145/3772318.3791358](#).
Category: CCF-A conference; top-tier venue in HCI.
- [C8] **CCFA** Fan, G.* , Liu, D., Pan, L., Zhang, R., & Guo, Q. (2026). Multi-LLM Persona Generation for Virtual Focus Groups in Software Engineering: A Controlled, Multi-Domain Study of Emotional Requirements Elicitation. *Proceedings of the ACM on Software Engineering (FSE 2026)*, ACM. DOI: [10.1145/3808098](#).
Category: CCF-A conference; top-tier venue in software engineering.
- [C9] **CCFA** Liu, D., Sabri, A.Q.M., Pan, L., & Fan, G.* (2026). Audit-of-Audits for the Web: Bayesian Meta-Evaluation that Yields Interval-Valued, Threshold-Aligned Fairness Claims. *In Proceedings of the ACM Web Conference 2026 (WWW 2026)*, ACM. DOI: [10.1145/3774904.3792974](#).
Category: CCF-A conference; top-tier venue in the Web.
- [C10] **CCFA** Fan, G.*[#], Liu, D.[#], & Pan, L. (2026). Mind the Gap: Predicting, Explaining and Reducing Time-to-First-Comment (Reply Gap) in Online Mental-Health Communities. *In Proceedings of the 40th AAAI Conference on Artificial Intelligence (AAAI 2026)*, AAAI Press. [Article #41184](#).
Category: CCF-A conference; top-tier venue in artificial intelligence.
- [C11] **CCFA** Fan, G.* , Liu, D., Pan, L., & Huang, Y. (2025). Creative Momentum Transfer: How Timing and Labeling of AI Suggestions Shape Iterative Human Ideation. *In Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI 2025)*. [Proceedings link](#).
Category: CCF-A conference; top-tier venue in artificial intelligence.

CCF-B Conferences (2 papers)

- [C12] Fan, G., Zhang, M., Zhao, M., & Pan, L. (2026). LePER: Label-Free Edge Polarity Reweighting for Heterophily. *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Barcelona, Spain, pp. 1716–1720. DOI: [10.1109/ICASSP55912.2026.11464154](#).
Category: CCF-B conference; flagship venue in signal processing. **Oral Presentation**.
- [C13] Fan, G.* , Liu, D., Pan, L., Zhang, R., & Sabri, A.Q.M. (2026). Consent Boundaries by Play: A CI-Grounded Micro-Game for Eliciting Stable Preferences and Diagnosing Opt-Out Friction in Web Consent. *In Proceedings of the European Conference on Computer-Supported Cooperative Work (ECSCW 2026)*.
Category: CCF-B conference; representative venue in CSCW.

Findings (1 paper)

- [C14] Fan, G., Liu, D., Guo, Q., Niu, X., & Pan, L. (2026). FairSynth: Instrument-Relative Evaluation and Calibration of LLM Survey Simulations for Measurement Fairness. *Findings of the Association for Computational Linguistics: EMNLP 2026*. **Accepted; forthcoming**.
Accepted; forthcoming. Findings is not listed in the CCF directory. [EMNLP main conference rank: A* \(ICORE2026\)](#).

Journal Papers

CCFA CCF-A Journals (1 paper)

- [J1] **CCFA** Fan, G.* , Liu, D., & Pan, L. (2026). Conformal@K: Distribution-Free Top-K Miss-Risk Control for Recommendation with Overlapping-Group and Two-Stage Guarantees. *ACM Transactions on Information Systems (TOIS)*, **44**(6), 1–42. DOI: [10.1145/3828550](#).
Category: CCF-A journal; CAS Zone 2; JCR Q1; **IF** $\approx$ 9.1; top-tier information retrieval journal.

CAS Zone 1 Top CAS Zone 1 Top Journals (5 papers)

- [J2] **CAS Zone 1 Top** Liu, D., & Fan, G.* (2026). “Fairness Labels” in Feeds: Process Disclosures, Provenance, and User Choice Among Short-Video Creators in Mainland China. *Social Media + Society*, **12**(3), Article 20563051261475922. DOI: [10.1177/20563051261475922](#).
Category: SSCI; CAS Zone 1 Top journal. **Role**: Corresponding author.

- [J3] **CAS Zone 1 Top** Fan, G.* , Liu, D., Zhang, R., & Pan, L. (2025). The impact of AI-assisted pair programming on student motivation, programming anxiety, collaborative learning, and programming performance: a comparative study with traditional pair programming and individual approaches. *International Journal of STEM Education*, **12**(1), 16. DOI: [10.1186/s40594-025-00537-3](https://doi.org/10.1186/s40594-025-00537-3).
Category: CAS Zone 1 Top in education; JCR Q1; IF ≈ 8.0 ; top-tier STEM education journal. Recognition: ESI Highly Cited Paper with 103 citations.
- [J4] **CAS Zone 1 Top** Liu, D., Fan, G.* , & Pan, L. (2026). Tool, Tutor, or Crutch?: A Grounded Theory of Cognitive Scaffolding and Offloading in AI-Assisted Programming Education. *International Journal of STEM Education*, **13**, 10. DOI: [10.1186/s40594-025-00592-w](https://doi.org/10.1186/s40594-025-00592-w).
Category: CAS Zone 1 Top in education; JCR Q1; IF ≈ 8.0 ; top-tier STEM education journal.
- [J5] **CAS Zone 1 Top** Fan, G., Sabri, A.Q.M.* , Rahman, S.S.A., & Pan, L. (2025). DynaKey-GNN: An efficient dynamic key-node multi-graph neural network for spatio-temporal traffic flow forecasting. *Engineering Applications of Artificial Intelligence*, **159**, 111757. DOI: [10.1016/j.engappai.2025.111757](https://doi.org/10.1016/j.engappai.2025.111757).
Category: CCF-C journal; CAS Zone 1 Top in computer science/engineering; JCR Q1; IF ≈ 9.0 ; representative AI applications journal.
- [J6] **CAS Zone 1 Top** Fan, G., Sabri, A.Q.M.* , Rahman, S.S.A., Pan, L., & Rahardja, S. (2025). Emerging Trends in Graph Neural Networks for Traffic Flow Prediction: A Survey. *Archives of Computational Methods in Engineering*, 1–45. DOI: [10.1007/s11831-025-10286-9](https://doi.org/10.1007/s11831-025-10286-9).
Category: CAS Zone 1 Top in engineering; JCR Q1; IF ≈ 12.9 ; top-tier computational engineering review journal.

CCF-B / C and Other International Journals (3 papers)

- [J7] Liu, D., Pan, L., Sabri, A.Q.M., & Fan, G.* (2026). From Solo Post to Shared Space: How a Public LLM Agent Reshapes Human-to-Human Conversation Structure on a Social Platform. *ACM Transactions on Social Computing*. Advance online publication. DOI: [10.1145/3844666](https://doi.org/10.1145/3844666).
Role: Corresponding author.
- [J8] Fan, G.* , Liu, D., & Huang, Y. (2025). Skim or Swim? Investigating AI-Generated Summaries, Trust, and Comprehension in Chinese Mobile Reading. *International Journal of Human-Computer Interaction*, 1–22. DOI: [10.1080/10447318.2025.2574513](https://doi.org/10.1080/10447318.2025.2574513).
Category: CCF-B journal; CAS Zone 2; JCR Q1; IF ≈ 6.0 ; representative HCI journal.
- [J9] Liu, D., Fan, G.* , Guo, Q., Zhang, R., & Pan, L. (2026). Textless, Tiny, and Private: Hashed n-grams with LLM-Teacher Distillation and Local Differential Privacy for Mental-Health Screening on Social Media. *IEEE Transactions on Computational Social Systems*. Early Access. DOI: [10.1109/TCSS.2026.3693692](https://doi.org/10.1109/TCSS.2026.3693692).
Category: CCF-C journal; CAS Zone 3; JCR Q2; IF ≈ 4.6 ; representative social computing journal.

Honors and Awards

- **Teaching Award** First Prize, University Teaching Innovation Competition, Taiyuan University of Science and Technology.
- **Teaching Award** Third Prize, Shanxi Provincial University Teaching Innovation Competition.
- **ACM CHI 2026 Honorable Mention Award** – *When Help Hurts: Verification Load and Fatigue with AI Coding Assistants*.
- **ESI Highly Cited Paper** (2025) – published in *International Journal of STEM Education*.
- **First Prize Paper and Best Presentation Award** (July 2025) – 9th China Computer Practice Education Academic Conference (CPEC 2025).

Research Projects

- **Participant, Shanxi Provincial Department of Science and Technology Laboratory Self-Initiated Project:** Key technologies for intelligent decision support based on multimodal big data in highway transportation. **Grant No.:** 202502ZZ0910029. **Period:** 2026-01 to 2027-12. **Total project budget:** RMB 950,000. **Status:** ongoing.
- **Participant, Shanxi Provincial Basic Research Program (General Program):** Meso-scale based massively parallel multi-agent simulation models for large-scale complex systems. **Grant No.:** 202203021221145.
- **Participant, Shanxi Higher Education Teaching Reform and Innovation Project:** Deepening undergraduate teaching evaluation reform: AI-enabled classroom quality evaluation and practice. **Grant No.:** J20250145.

Academic Service

Program Committee

- **The Web Conference (WWW) 2026** – Web4Good Special Track (CCF A)

Reviewer

- **CHI 2026** – ACM Conference on Human Factors in Computing Systems (CCF A)
- **ICML 2026** – International Conference on Machine Learning (CCF A)
- **Engineering Applications of Artificial Intelligence** (EAAI, CAS Zone 1 Top)
- **International Journal of STEM Education** (CAS Zone 1 Top)
- **Information, Communication & Society** (SSCI Q1 Top)